

The simplest global path

Two-pin global routing often starts with an L: travel along one axis, then the other

The idea

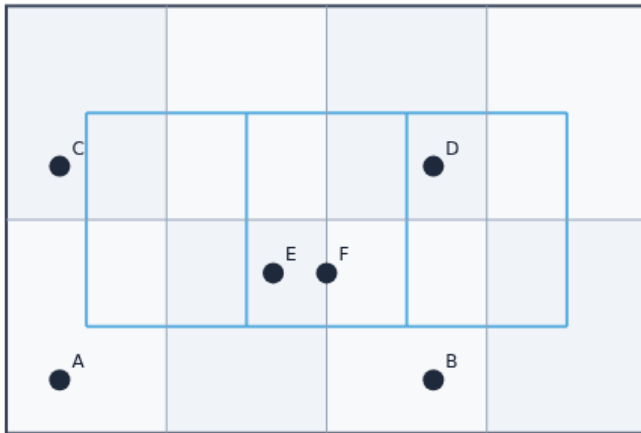
- Prefer VH swaps the order
- Emit the list of GCells visited and convert consecutive pairs to edges with `path_to_edges`
- A path from zero comma zero to two comma one in HV uses three edges

L-shape

L-PATTERN ROUTE

L-shape

Step 1 / 5 · idea · module02-01-pattern-l-route



Explanation

Two-pin nets use one bend: HV (horizontal first) or VH.

- Manhattan
- One corner

Pattern router

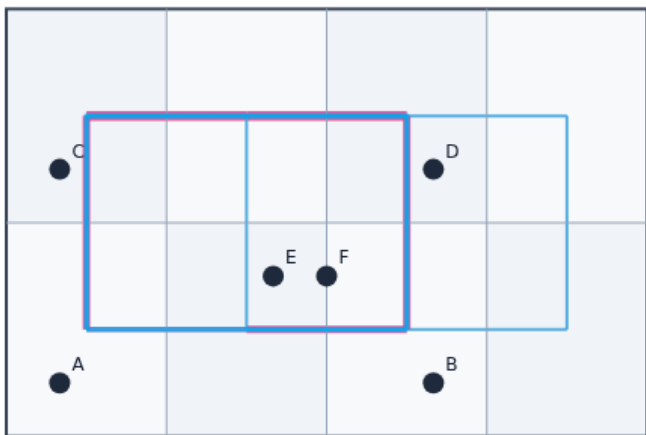
L-shape

Route L-HV

L-PATTERN ROUTE

Route L-HV

Step 2 / 5 · hv · module02-01-pattern-l-route



Explanation

Spread placement L-HV yields documented overflow.

- All six nets
- Shared corridors

total≈2

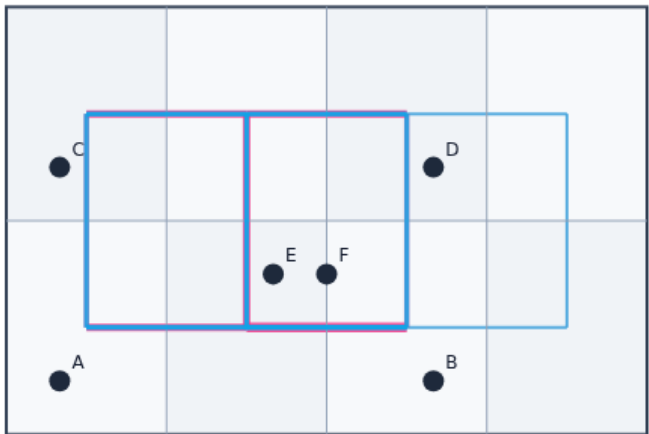
Route L-HV

Route L-VH

L-PATTERN ROUTE

Route L-VH

Step 3 / 5 · vh · module02-01-pattern-l-route



Explanation

Swapping bend order changes which edges saturate.

- Same pins
- Different edges

VH total=1

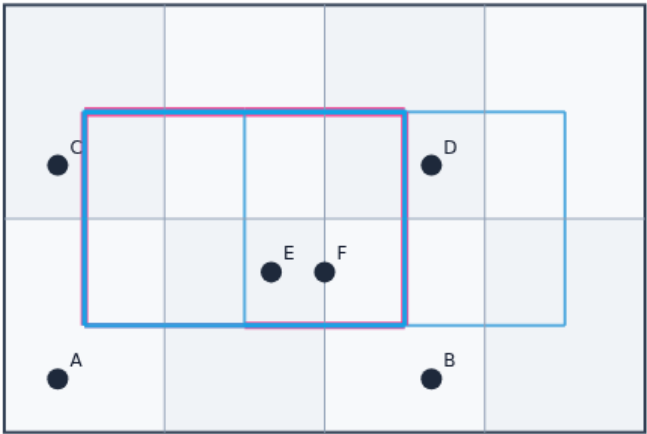
Route L-VH

Edge usage

L-PATTERN ROUTE

Edge usage

Step 4 / 5 · edges · module02-01-pattern-l-route



Explanation

Thicker edges show usage; red dash marks overflow.

- usage vs cap
- max/count

Heat on edges

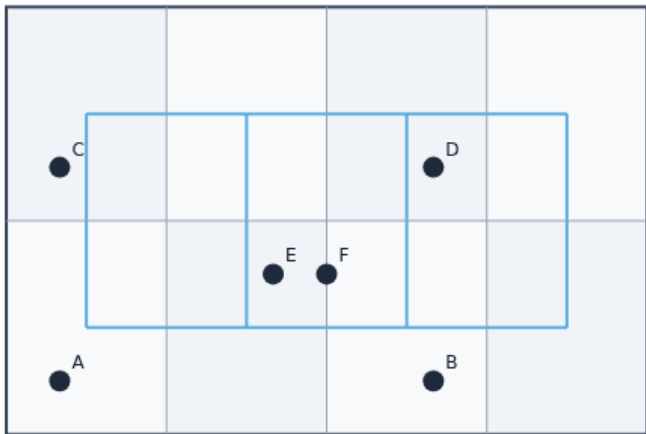
Edge usage

Move pins

L-PATTERN ROUTE

Move pins

Step 5 / 5 · move · module02-01-pattern-l-route



Explanation

Moving cells changes terminal GCells and L paths.

- Learner state
- Check scores routes

Browser lab next

Move pins

Browser lab track

EDA Algorithms Platform

HomeTools

HomeToolsL-pattern routing

INTERACTIVE TOOL

L-pattern routing

Route nets with horizontal-then-vertical or vertical-then-horizontal L-shapes.

Your job: click Route L-HV or L-VH on spread starter. HV total overflow $\approx$ 2; VH $\approx$ 1.

Reset workspace

Challenges0 / 8 cleared

Pick one [Intro] Route L-HV

Pick a challenge and click **Start**. Move cells, route, then **Check**. Reveal golden is study-only.

Start

Show hint

Check

Next challenge

Stop checking

Idle

A

B

C

D

E

F

←

→

↑

↓

Reset to starter

Reveal golden (study)

Route L-HV

Route L-VH

Route maze

Rip-up reroute

Graph

Reference / metrics

selected: A @ GCell (0,0)

routes: (none — click Route)

overflow total: 0.00 · max: 0.00 · count: 0

edge capacity: 2

HPWL: 34.0

view: your placement

Browser lab starter

learn_global_routing — 01 pattern | route

Implement track

- Implement ``l_route(a, b, prefer)`` and ``path_to_edges``
- Route net A-B on `tiny_gr` and print the edge list
- Match the browser overlay

Pitfalls

- Skipping duplicate GCells at the bend
- Returning directed edges inconsistently, always normalize undirected keys
- Forgetting that L routes ignore existing congestion until sequential routing

Your turn

- Ship Track A L-routes and clear browser challenges
- Next: Z-shape patterns when you want a third segment